

SEMESTER END EXAMINATION (SEE) B. E. 7th SEMESTER – DECEMBER 2010

IS 402 – OBJECT ORIENTED ANALYSIS AND DESIGN

Time: 3 Hrs

Answer All Questions

Max Marks: 100

1.	a)	Explain briefly the five views for describing the software intensive system architecture.	10
	b)	Consider a retail store called "XYZ COMPANY". It takes orders from its customers. The customers may be located at different parts of the country. Once the customer orders for product(s), they are checked against the inventory present in their warehouse. If present, an invoice is generated through a set of transactions. Finally, the products are packed and kept ready for delivery through shipment. The company provides free shipment for its customers. Suppose XYZ COMPANY is interested in automating the complete ordering process, identify the classes required along with suitable members and represent them using appropriate UML diagram.	04
	c)	i) Mr. James has a CPU. Generally, a CPU may be associated with no or many controllers depending on the usage of disk drives. Mr. James has four disk drives and wants an efficient controller to control these drives. He opts for a special type of controller called SCSI controller which has ability to control a minimum of one disk drive to a maximum of four disk drives. Write a UML diagram to model Mr. James system's components specified in the given scenario. ii) Consider the following class Skelton. Analyses and write an equivalent UML diagram for the same. <pre> public class Sale { public void updatePriceFor(ProductDescription description) { Money basePrice = description.getPrice(); //... } // ...}; </pre>	02
	d)	What is a stereotype? What are three ways of representing the stereotype?	02
2.	a)	i) Write a class diagram for the following object diagram. <pre> classDiagram class Spain["Spain: Country"] { Name := Spain } class France["France: Country"] { Name := France } class Belgium["Belgium: Country"] { Name := Belgium } Spain -- France : Borders France -- Belgium : Borders </pre> ii) An undirected graph consists of a set of vertices and a set of edges. Edges connect pair of vertices. Prepare a class model to describe such an undirected graph. Note that the model should capture only the structure of the graph i.e. connectivity.	02
	b)	Categorize the following into different types of UML relationships i) CPU and display are parts of computer ii) A user can login as admin or guest. iii) A bank has one or more accounts where each account can be checking, savings or money market. Note: Draw suitable class diagrams to illustrate the above relationship. Indicate association and multiplicities.	04
	c)	Explain briefly the dependency relationships among	08
		i. Class diagrams (any five) ii. Packages iii. Use cases.	
	d)	The following code illustrates a template class Set fragment that can be used to define behaviour for sets in general. <pre> class Set <T> { public: void insert (T newElement); void remove (T anElement); }; </pre> Write a UML template class to illustrate the model of instantiation of Set <Employee> employeeSet;	03

